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United States Patent	12397152
Kind Code	B2
Date of Patent	August 26, 2025
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Systems and methods for implanting a medical device

Abstract

In at least one embodiment, a system and method for implanting an implantable medical device (IMD) within a patient may include an IMD including a housing and an attachment member, and a delivery catheter including a tethering snare that is configured to be selectively extended out of the delivery catheter and retracted into the delivery catheter. In at least one embodiment, a system and method for implanting an implantable medical device (IMD) within a patient may include an IMD including a housing and an attachment member, wherein the attachment member includes a central passage connected to a connection chamber, and a delivery catheter including first and second tethers that may be moved outwardly from and retracted into the delivery catheter.

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Appl. No.:	18/375702
Filed:	October 02, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240024662 A1	Jan. 25, 2024

Related U.S. Application Data

continuation parent-doc US 17316434 20210510 US 11786723 child-doc US 18375702
division parent-doc US 16189891 20181113 US 11027119 20210608 child-doc US 17316434
division parent-doc US 14481818 20140909 US 10716931 20200721 child-doc US 16189891

Publication Classification

Int. Cl.: **A61N1/00** (20060101); **A61N1/05** (20060101); **A61N1/372** (20060101); **A61N1/375** (20060101); A61B17/00 (20060101); A61B17/34 (20060101); A61M25/00 (20060101)

U.S. Cl.:

CPC **A61N1/05** (20130101); **A61N1/37205** (20130101); **A61N1/3756** (20130101); A61B2017/00473 (20130101); A61B17/3468 (20130101); A61M25/0082 (20130101); A61N2001/058 (20130101)

Field of Classification Search

CPC: A61N (1/05); A61N (1/37205); A61N (1/3756); A61N (2001/058); A61B (17/3468); A61B (2017/00473); A61M (25/0082)

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Background/Summary

PRIORITY CLAIM (1) This application is a continuation of U.S. patent application Ser. No. 17/316,434, filed on May 10, 2021, which is a Divisional application of U.S. patent application Ser. No. 16/189,891, filed on Nov. 13, 2018 (now issued as U.S. Pat. No. 11,027,119), which is a Divisional application of U.S. patent application Ser. No. 14/481,818, filed on Sep. 9, 2014 (now issued as U.S. Pat. No. 10,716,931), and these applications are incorporated herein by reference in their entirety to provide continuity of disclosure.

BACKGROUND

- (1) Embodiments of the present disclosure generally relate to implantable medical devices, and, more particularly, to systems and methods for implanting a medical device.
- (2) Numerous medical devices exist today, including but not limited to electrocardiographs (“ECGs”), electroencephalographs (“EEGs”), squid magnetometers, implantable pacemakers, implantable cardioverter-defibrillators (“ICDs”), neurostimulators, electrophysiology (“EP”) mapping and radio frequency (“RF”) ablation systems, and the like. Implantable medical devices (hereinafter generally “implantable medical devices” or “IMDs”) are configured to be implanted within patient anatomy and commonly employ one or more leads with electrodes that either receive or deliver voltage, current or other electromagnetic pulses (generally “energy”) from or to an organ or tissue for diagnostic or therapeutic purposes.
- (3) Typically, an intra-cardiac IMD is introduced into the heart through a catheter. However, trans-catheter delivery of an entire IMD within a heart typically requires specialized tools. Often, the specialized tools are complex and may be difficult to manipulate and operate.
- (4) In general, an IMD may be connected to a delivery system in a docked state, in which the IMD is securely attached to the delivery system. In the docked state, the catheter may be operated to guide the IMD to an implant site. Once the IMD is proximate to the implant site, because the IMD is securely connected to the catheter, the catheter may be used to torque the IMD into patient tissue.
- (5) Once the IMD is secured into patient tissue, the IMD may be moved into a tethered state with respect to the catheter. In the tethered state, the catheter separates from the IMD, but remains connected thereto. In the tethered state, an implanting physician may test the IMD to make sure that the IMD is securely and electrically connected to patient tissue at a desired location. If the physical and/or electrical connection between the IMD and the patient tissue is less than optimal, the IMD may be re-docked to the catheter so that the IMD may be moved to a better position for implantation.
- (6) Once the implanting physician is satisfied with the location of the IMD within patient anatomy, the IMD is transitioned from the tethered state to a release state. In the release state, the IMD disconnects from the catheter.

(7) However, known systems and methods for releasing an IMD from a catheter are often susceptible to spontaneous release, in which the IMD inadvertently releases from the catheter. Further, known release systems and methods may not release the IMD smoothly and easily from the catheter. Also, known release systems and methods may malfunction and fail to release the IMD from the catheter.

SUMMARY

(8) Certain embodiments provide a system for implanting an IMD within a patient. The system may include an IMD including a housing and an attachment member. The system may also include a delivery catheter including a tethering snare that is configured to be selectively extended out of the delivery catheter and retracted into the delivery catheter. The tethering snare may be configured to fit over at least a portion of the attachment member in a fully extended position. Further, the tethering snare may be configured to securely tether to the attachment member in a retracted position. The tethering snare may be operable to retrievably connect the IMD to the delivery catheter, and release the IMD from the delivery catheter.

(9) In at least one embodiment, the tethering snare forms a loop that extends out of the delivery catheter. The loop is configured to fit over the portion of the attachment member in the fully extended position. The loop is configured to constrict around the portion of the attachment member in the retracted position. A size of the loop may increase when the tethering snare is extended out of the delivery catheter. The size of the loop may decrease when the tethering snare is retracted into the delivery catheter.

(10) The attachment member may include a neck extending from the housing, and an expanded head connected to the neck. The tethering snare may be configured to fit over the expanded head in the fully extended position, and securely constrict around the neck proximate to the expanded head in the retracted position. In at least one embodiment, the neck is pivotally secured to the housing. The attachment member may also include at least one torque recess, and the delivery catheter may include at least one torque key. The torque recess(es) is configured to securely mate with the torque key(s) in a docked state.

(11) Certain embodiments of the present disclosure provide a method for implanting an IMD within a patient. The method may include extending a tethering snare out of a delivery catheter to fit over a portion of an attachment member of the IMD, moving the extended tethering snare over the portion of the attachment member to a connecting position, retracting the tethering snare into the delivery catheter to securely connect the tethering snare to the IMD at the connecting position, and releasing the IMD from the delivery catheter by extending the tethering snare out of the delivery catheter so that the tethering snare disengages from the connecting position, and removing the tethering snare from the attachment member.

(12) Certain embodiments of the present disclosure provide a system for implanting an implantable medical device (IMD) within a patient. The system may include an IMD including a housing and an attachment member. The attachment member may include a central passage connected to a connection chamber. The system may also include a delivery catheter including first and second tethers that may be moved outwardly from and retracted into the delivery catheter. The first tether may include a protuberance at a distal end. The protuberance is sized to pass into the central passage. The protuberance and the second tether are configured to be lodged into one or both of the central passage and the connection chamber to securely tether the IMD to the delivery catheter.

(13) The second tether may include an elongated interfering segment that is configured to be retracted into the delivery catheter and removed from the central passage. The protuberance may be removed from the central passage in response to the elongated interfering segment being removed from the central passage. In at least one embodiment, the second tether may be featureless and devoid of any protuberance.

(14) A protuberance diameter of the protuberance may be less than a passage diameter of the central passage. A tether diameter of the second tether may be less than either of the passage

diameter and the protuberance diameter. A combined diameter of the tether diameter and the protuberance diameter may be greater than the passage diameter.

(15) Certain embodiments of the present disclosure provide a method for implanting an IMD within a patient. The method may include securing the IMD to a delivery catheter by positioning a first tether within a central passage of an attachment member of the IMD. The positioning the first tether within the central passage prevents a protuberance of a second tether from passing into the central passage. The method may also include releasing the IMD from the delivery catheter by removing the first tether from the central passage. The removing the first tether from the central passage allows the protuberance to be removed from the central passage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a perspective view of a delivery system for delivering an implantable medical device (IMD) into a patient, according to an embodiment of the present disclosure.
- (2) FIG. 2A illustrate a perspective view of a distal portion of a delivery system and an IMD, according to an embodiment of the present disclosure.
- (3) FIG. 2B illustrates a perspective view of an IMD sheath extended distally along a guide shaft, according to an embodiment of the present disclosure.
- (4) FIG. 3A illustrates a perspective view of a delivery system disconnected from an IMD, according to an embodiment of the present disclosure.
- (5) FIG. 3B illustrates a perspective view of a distal end of a delivery catheter, according to an embodiment of the present disclosure.
- (6) FIG. 3C illustrates a perspective view of a delivery catheter tethered to an IMD, according to an embodiment of the present disclosure.
- (7) FIG. 3D illustrates a lateral view of a delivery catheter tethered to an IMD, according to an embodiment of the present disclosure.
- (8) FIG. 4 illustrates a perspective top view of a proximal end of an IMD, according to an embodiment of the present disclosure.
- (9) FIG. 5 illustrates a lateral view of a proximal end of an IMD, according to an embodiment of the present disclosure.
- (10) FIG. 6 illustrates a lateral view of a tethering snare extending from a distal end of a delivery catheter, according to an embodiment of the present disclosure.
- (11) FIG. 7 illustrates a lateral view of a tethering snare extending from a distal end of a delivery catheter in an expanded state, according to an embodiment of the present disclosure.
- (12) FIG. 8 illustrates a lateral view of a tethering snare extending from a distal end of a delivery catheter in a constricted state, according to an embodiment of the present disclosure.
- (13) FIG. 9 illustrates a lateral view of a delivery catheter unsecured to an IMD, according to an embodiment of the present disclosure.
- (14) FIG. 10 illustrates a lateral view of a delivery catheter securely tethered to an IMD, according to embodiment of the present disclosure.
- (15) FIG. 11 illustrates a flow chart of a method of implanting an IMD at an implant site, according to an embodiment of the present disclosure.
- (16) FIG. 12 illustrates a perspective top view of an attachment member of an IMD, according to an embodiment of the present disclosure.
- (17) FIG. 13 illustrates a cross-sectional view of an attachment member of an IMD through line 13-13 of FIG. 12, according to an embodiment of the present disclosure.
- (18) FIG. 14 illustrates a lateral view of a protuberance tether of a delivery catheter, according to an embodiment of the present disclosure.

(19) FIG. 15 illustrates a lateral view of a locking tether of a delivery catheter, according to an embodiment of the present disclosure.

(20) FIG. 16A illustrates a cross-sectional view of a protuberance tether passing into a connection chamber of an attachment member of an IMD, according to an embodiment of the present disclosure.

(21) FIG. 16B illustrates an axial cross-sectional view of a protuberance within a central passage of an attachment member, according to an embodiment of the present disclosure.

(22) FIG. 16C illustrates an axial cross-sectional view of a protuberance within a central passage of an attachment member, according to an embodiment of the present disclosure.

(23) FIG. 17 illustrates a cross-sectional view of a protuberance tether and locking tether securely lodged within a connection chamber of an attachment member of an IMD, according to an embodiment of the present disclosure.

(24) FIG. 18 illustrates a perspective bottom view of a protuberance of a protuberance tether and a locking tether securely lodged within a connection chamber of an attachment member of an IMD, according to an embodiment of the present disclosure.

(25) FIG. 19 illustrates a flow chart of a method of implanting an IMD at an implant site, according to an embodiment of the present disclosure.

(26) FIG. 20 illustrates a simplified view of an IMD in electrical communication with at least three leads implanted into a patient's heart, according to an embodiment.

DETAILED DESCRIPTION

(27) Embodiments of the present disclosure provide systems and methods of attaching an IMD to a catheter, and disconnecting the IMD from the catheter in an intuitive, straightforward, and easy manner. The IMD may be any one of various types of implantable devices, such as, for example, an implantable pacemaker, implantable cardioverter-defibrillator (“ICD”), defibrillator, cardiac rhythm management (“CRM”) device, neurostimulator, or the like.

(28) In at least one embodiment, the IMD may include a leadless cardiac pacemaker that may be enclosed in a hermetic housing or can that may be positioned on the inside or outside of a cardiac chamber. The pacemaker may have two or more electrodes located within, on, or near the housing, for delivering pacing pulses to muscle of the cardiac chamber and optionally for sensing electrical activity from the muscle, and for bidirectional communication with at least one other device within or outside the body. The housing may contain a primary battery to provide power for pacing, sensing, and communication, for example bidirectional communication. The housing may optionally contain circuits for sensing cardiac activity from the electrodes. The housing may contain circuits for receiving information from at least one other device via the electrodes and may contain circuits for generating pacing pulses for delivery via the electrodes. The housing may optionally contain circuits for transmitting information to at least one other device via the electrodes and may optionally contain circuits for monitoring device health. The housing may contain circuits for controlling these operations in a predetermined manner.

(29) FIG. 1 illustrates a perspective view of a delivery system **100** for delivering an IMD **102** into a patient, according to an embodiment of the present disclosure. The delivery system **100** may include an IMD sheath **104**, a guide catheter **111**, an introducer sheath **107**, a handle **108**, a deflection knob **110**, a tether shuttle **112**, and flush ports **114a**, **114b**, and **114c**. The deflection knob **110** may be used to steer and guide the catheter **111** during implantation and/or removal of the IMD **102**. The flush ports **114a**, **114b**, and **114c** may be used to flush saline or other fluids through the catheter **111**. The introducer sheath **107** may be advanced distally over the catheter **111** to provide additional steering and support for the catheter **111** during implantation and to surround the IMD **102** as it is introduced through a trocar or introducer into a patient.

(30) FIG. 2A illustrate a perspective view of a distal portion of a delivery system **200** and an IMD **202**, according to an embodiment of the present disclosure. The IMD **200** may include a helix **203** that may be used to attach the IMD **200** to tissue of a patient. The IMD **202** may include an

attachment member that is configured to removably connect to a docking cap **218** of a catheter **206**. An IMD sheath **204** is shown pulled back proximally along the catheter **206** and a guide shaft **211** to expose the IMD **202** and the helix **203**.

(31) FIG. 2B illustrates a perspective view of the IMD sheath **204** extended distally along the guide shaft **211** to cover the catheter **206**, the IMD **202**, and the helix **203**, according to an embodiment of the present disclosure. The extended IMD sheath **204** protects patient tissue from sharp edges of the helix **203** during implantation. Referring to FIGS. 2A and 213, when the IMD sheath **204** is pulled back proximally, as shown in FIG. 2A, the IMD **202** is in an exposed, delivery configuration. When the IMD sheath **204** is advanced distally to protect the IMD **202** and the helix **203**, as shown in FIG. 213, the IMD **202** is in a protected, advancement configuration.

(32) FIG. 3A illustrates a perspective view of a delivery system **400** disconnected from an IMD **402**, according to an embodiment of the present disclosure. The delivery system **400** may include the IMD **402** when the IMD **402** is connected to the delivery system **400**. The IMD **402** may include a helix **403** and an attachment member **424**, such as a docking button, cap, stud, ridge, ledge, rim or the like.

(33) The delivery system **400** may include a delivery catheter **407** that may include an IMD sheath **404**, a catheter shaft **406**, a docking cap **418**, and a tethering snare **422**. The tethering snare **422** may be or include one or more wires, shafts, tubes, cords, ropes, strings, or other similar structures that may extend throughout the catheter shaft **406**. In at least one embodiment, the tethering snare **422** may include a shape memory material, such as nitinol. In other embodiments, the tethering snare **422** may include stainless steel wires or braids. As shown in FIG. 3A, the IMD **402** is disconnected from the docking cap **418** of the delivery catheter **407**.

(34) FIG. 3B illustrates a perspective view of a distal end of the delivery catheter **407**, according to an embodiment of the present disclosure. The tethering snare **422** is configured to be moved into and out of the delivery catheter **407** to expand its size. For example, the tethering snare **422** may be extended out of the docking cap **418** to expand a diameter **423** of a loop **425** of the tethering snare **422**, as described in detail below. In at least one embodiment, the IMD sheath **404** of the delivery catheter **407** may be moved over the tethering snare **407** over the catheter shaft **406** in the direction of arrow **411** in order to reduce the size of the tethering snare **422**. As the IMD sheath **404** retreats back over the catheter shaft **406** in the direction of **411'**, the size of the tethering snare **422** increases. The expanded tethering snare **422** is used to loop over and snare the attachment member **424** (shown in FIG. 3A). Once the attachment member **424** is snared, the tethering snare **422** is pulled back into the delivery catheter **407** (such as by an operator pulling the proximal ends of the tethering snare **422** into the delivery catheter **407**, and/or the IMD sheath **404** being moved over the tethering snare **422** in the direction of arrow **411**) in order to securely tether the IMD to the delivery catheter **407**, as described in detail below.

(35) In at least one embodiment, the delivery catheter **407** may include a locking sheath (which may be or form part of the IMD sheath **404**) that collapses the tethering snare **422**. For example, the tethering snare **422** may be formed of a resilient material, such as nitinol. As such, the tethering snare **422** springs back to an at-rest shape when the locking sheath is removed from the tethering snare **422**.

(36) FIG. 3C illustrates a perspective view of the delivery catheter **407** tethered to the IMD **402**, according to an embodiment of the present disclosure. FIG. 3D illustrates a lateral view of the delivery catheter **407** tethered to the IMD **402**. Referring to FIGS. 3C and 3D, the tethering snare **422** is positioned underneath a securing member **427**, such as a head, stud, block, barb, or the like, of the attachment member **424** so that the loop **425** catches, snags, or snares thereon. After the loop **425** snares onto the attachment member **424**, the tethering snare **424** is drawn into the delivery catheter **407** so that the tethering snare **424** securely tethers the IMD **402** to the delivery catheter **407**.

(37) The docking cap **418** of the delivery catheter may include a torque slot that is sized and

configured to mate with a torque key **432** located on a proximal end of the pacemaker IMD **402**. The torque slot may be coupled to a torque shaft, which may run the length of the delivery catheter extending into the handle (not shown). The torque key may be a “male” key and the torque slot may be a “female” key, or vice versa. The torque key and the torque slot may include any number of shapes, such as square, rectangle, triangle, pentagon, hexagon, cross, “X”, and the like, so long as the key fits within and can apply rotational torque to the slot. Once the tethering snare **422** securely tethers to the attachment member **424**, the tethering snare **422** may be pulled proximally to pull the attachment feature **424** and therefore the IMD **402** towards the delivery catheter **407** and to attach the IMD **402** to the delivery catheter **407**, thereby engaging the torque slot with the torque key **432**.

(38) Aspects of the delivery catheter **407** and the IMD **402** may be further described in United States Patent Application Publication No. 2014/0074114, entitled “Delivery Catheter Systems and Methods,” which is hereby incorporated by reference in its entirety.

(39) FIG. 4 illustrates a perspective top view of a proximal end of an IMD **500**, according to an embodiment of the present disclosure. The IMD **500** may include an attachment member **502** connected to a housing or can **504**. The attachment member **502** may be or include a docking button, for example, that is configured to removably connect to a delivery catheter.

(40) The attachment member **502** may include a neck **506** that is pivotally secured to a proximal end of the housing **504** through a central guide pin that is rotatably secured within reciprocal channels formed through a collar **508** that extends upwardly from the housing **504**. The rotatable connection between the pin and the channels allows the attachment member **502** to pivot in the directions of arc **510** about an axis defined by the central guide pin. Alternatively, the housing **504** may include the central guide pin, while the neck **506** includes a channel that receives the central guide pin.

(41) The collar **508** includes an outer wall **512** that defined a cavity **514** in which the neck **506** is positioned. The outer wall **512** limits the pivotal movement of the neck within the cavity **514**. Alternatively, the housing **504** may not include the collar **508**.

(42) FIG. 5 illustrates a lateral view of the proximal end of the IMD **500**, according to an embodiment of the present disclosure. As shown in FIG. 5, the neck **506** may include the central guide pin **516** having ends that are pivotally retained within channels formed through brackets **518** extending from the collar **508**. Accordingly, the attachment member **502** may pivot in the directions denoted by arc **510** about the central longitudinal axis **520** of the central guide pin **516**. The attachment member **502** may pivot in order to allow the IMD **500** to bend, pivot, or otherwise articulate as the IMD **500** is navigated to an implant site. Accordingly, the IMD **500** may be moved with greater ease and precision through vasculature of a patient. Alternatively, the attachment member **502** may be fixed in position, and may not be configured to pivot with respect to the housing **504**.

(43) Referring to FIGS. 4 and 5, the attachment member **502** may also include an expanded head **530** connected to an end of the neck **506** that is opposite from an end **534** that pivotally secures to the housing **504**. The expanded head **530** may have a diameter that is greater than that of the neck **506**. The head **530** may include one or more torque recesses **534**, such as divots, cut-outs, cored-out areas, slots, slits, or the like, formed therethrough. The torque recesses **534** may be scalloped areas that allow for increased torque transfer. The torque recesses **534** are configured to mate with reciprocal torque keys within a delivery catheter, as described above. Alternatively, the head **530** may include one or more torque keys, while the delivery catheter includes the torque recesses **534**.

(44) Each torque recess **534** may be formed through the head **530** from a top surface **536** to a bottom surface **538**. Alternatively, each torque recess **534** may be formed from the top surface **536** to an intermediate area above the bottom surface **538**. More or less torque recesses **534** may be used. For example, the head **530** may include a single torque recess, or three or more regularly spaced torque recesses.

(45) FIG. 6 illustrates a lateral view of a tethering snare **600** extending from a distal end **602** of a delivery catheter **604**, according to an embodiment of the present disclosure. The tethering snare **600** is an example of such as shown and described with respect to FIGS. 3A-3D. The tethering snare **600** may be or include a loop **606** (which may include a single loop or a plurality of loops) of wire, string, or the like that extends outwardly from an internal passage **608** of the delivery catheter **604**. Ends of the loop **606** may be operatively connected to a control device, such as the shuttle **112** shown in FIG. 1, a control knob, joystick, button(s), and/or the like of an IMD delivery system, similar to the system **100** shown in FIG. 1. The loop **606** may be drawn into the internal passage **608** to decrease its size, or pushed outwardly from the internal passage **608** to increase its size, such as through movement of the tethering snare **600** into and out of the delivery catheter **604**, and/or through movement of the delivery catheter **604** or sheath in relation to the tethering snare **600**.

(46) The tethering snare **600** may be a single layer of material, such as string or wire, or multiple layers of material. For example, the tethering snare **600** may be a braided or woven piece formed through multiple strings, wires, or the like.

(47) FIG. 7 illustrates a lateral view of the tethering snare **600** extending from the distal end **602** of the delivery catheter **604** (which may be or include an IMD sheath) in an expanded state, according to an embodiment of the present disclosure. As shown in FIG. 7, the tethering snare **600** has been pushed, extended, or otherwise moved outwardly from the delivery catheter **604** in the direction of arrow **610**. For example, the delivery catheter **604** (or an IMD sheath of the delivery catheter **604**) may be pushed over the tethering snare **600** so that the tethering snare **600** retracts, contracts, recedes, retreats, or the like back into the delivery catheter **604**. Conversely, the delivery catheter **604** may be retracted in relation to the tethering snare **600** so that the tethering snare **600** extends outwardly from the sheath. With increased urging of the tethering snare **600** in the direction of arrow **610**, a diameter **612** of the loop **606** expands, enlarges, or otherwise increases.

(48) FIG. 8 illustrates a lateral view of the tethering snare **600** extending from the distal end **602** of the delivery catheter **604** in a constricted state, according to an embodiment of the present disclosure. When the tethering snare **614** is drawn back into the internal passage **608** of the delivery catheter **604** in the direction of arrow **614**, the diameter **612** of the loop decreases.

(49) FIG. 9 illustrates a lateral view of the delivery catheter **604** unsecured to the IMD **500**, according to an embodiment of the present disclosure. As shown in FIG. 9, the diameter **612** of the loop **606** of the tethering snare **600** is greater than the diameter **700** of the head **530** of the IMD **500**. As such, the tethering snare **600** may pass over the head **530**. In order to securely connect or tether the delivery catheter **604** to the IMD **500**, a portion of the loop **606** may be positioned underneath the bottom surface **538** of the head **530**. An operator may then draw the tethering snare **600** up into delivery catheter **604** (such as by pulling the tethering snare **600** into the delivery catheter **604**, or sliding the delivery catheter **604** over the tethering snare **600**) so that the diameter **612** is less than the diameter **700**. In this manner, the tethering snare **600** is prevented from disconnecting from the attachment member **502**. The loop **606** may then be further retracted so that it snugly wraps around the neck **506** of the attachment member **502** underneath the bottom surface **538** of the head **530**.

(50) FIG. 10 illustrates a lateral view of the delivery catheter **604** securely tethered to the IMD **500**, according to embodiment of the present disclosure. As shown, the tethering snare **600** is securely positioned at a connecting position **605**, in which the retracted tethering snare **600** is wrapped around the neck **506** underneath the bottom surface **538** of the head **530**. As such, the tethering snare **600** may constrict around the neck **506** at the connecting position **605**. Because the diameter **612** of the loop **606** that extends outwardly from the delivery catheter **604** is less than the diameter **700** of the head **530**, the tethering snare **600** remains secured to the IMD **500**. As such, the IMD **500** is securely tethered to the delivery catheter **604**.

(51) In order to securely dock the IMD **500** to the delivery catheter **604**, the tethered IMD **500** may be drawn up into the internal passage **608** of the delivery catheter **604** in the direction of arrow **610**.

Within the internal passage **608**, the torque recesses **534** may mate with reciprocal torque keys within the delivery catheter **604**.

(52) In order to release the IMD **500** from the delivery catheter **604**, the tethering snare **600** is pushed outwardly from the delivery catheter **604** so that the diameter **612** of the loop **606** exceeds the diameter **700** of the head **530**, as shown in FIG. **9**. As such, the loop **606** may pass over the head **530** in the direction of arrow **702** to disconnect the delivery catheter **604** from the IMD **500**.

(53) FIG. **11** illustrates a flow chart of a method of implanting an IMD at an implant site, according to an embodiment of the present disclosure. The process begins at **800**, in which a delivery catheter containing an IMD is guided to an implant site. Once at the implant site, the IMD is pushed out from the delivery catheter and fixed to the implant site at **802**. Then, at **804**, the delivery catheter is moved away from the implant site while remaining tethered to the IMD.

(54) When the IMD is fixed to the implant site and tethered to the delivery catheter, the IMD is then tested at **806** to determine whether the IMD is properly physically and electrically connected to the implant site. At **808**, it is determined whether the IMD is properly affixed to the implant site. If not, the method proceeds to **810**, in which the IMD is removed from the implant site and retracted back into the delivery catheter and docked thereto. The process then returns to **800**.

(55) If, however, the IMD is properly affixed to the implant site, at **812**, a tethering snare (which tethers the delivery catheter to the IMD) is pushed out of the delivery catheter to increase a diameter of a loop of the tethering snare. At **814**, it is determined whether the loop extending from the delivery catheter is greater than a head of an attachment member of the IMD. If not, the process returns to **812**. If, however, the loop extending from the delivery catheter is greater than the head of the attachment member, the tethering snare is removed from the IMD at **816**, such as by slipping the loop over and off the attachment member, thereby releasing the IMD from the delivery catheter.

(56) As described above, embodiments of the present disclosure provide a system and method for securing and releasing an IMD from a delivery catheter. The tethering snare may also be used to retrieve an IMD from an implanted position. The delivery catheter may include a tethering snare that may be extend out of, and retracted into, a locking sheath, for example, of the delivery catheter. The tethering snare is configured to expand to fit over the attachment member of the IMD, and then be constricted and tightened to securely tether to the attachment member.

(57) It has been found that the embodiments described with respect to FIGS. **4-11** are less susceptible to changes in length (compared to known systems) that would otherwise occur due to changes in temperature, deflection, torsion, and the like. Additionally, the embodiments incorporate a delivery and retrieval system into a single system. For example, the tethering snare may be used to release the attachment member of the IMD, and securely re-connect the IMD to the delivery catheter. Further, the embodiments described with respect to FIGS. **4-11** are not susceptible to inadvertent, spontaneous release, for example.

(58) FIG. **12** illustrates a perspective top view of an attachment member **900** of an IMD, according to an embodiment of the present disclosure. The attachment member **900** may be a part of the IMD. In at least one embodiment, the attachment member **900** may be a proximal end of a housing or can of the IMD. In at least one other embodiment, the attachment member **900** may fixedly or pivotally connect to the housing or can of the IMD through a neck, such as described above with respect to FIGS. **4** and **5**.

(59) The attachment member **900** may include a main body **902** having one or more torque recesses **904**, as described above. A central passage **906** is formed through a top surface **908** of the main body **902**. The central passage **906** may be aligned with and about a central axis **910** of the attachment member **900**. The central passage **906** connects to an internal connection chamber formed within the main body **902**.

(60) FIG. **13** illustrates a cross-sectional view of the attachment member **900** of the IMD through line **13-13** of FIG. **12**, according to an embodiment of the present disclosure. As shown, the central passage **906** connects to the internal connection chamber **912**. The connection chamber **912** may be

wider and larger than the central passage **906**. For example, the connection chamber **912** may have a diameter that is twice that of the central passage **906**. The central passage **906** may connect to the connection chamber **912** through outwardly flared walls **914**. The flared walls **914** may angle outwardly and down from the central passage **906** to the connection chamber **912**. Alternatively, the flared walls **914** may be flat walls that are perpendicular to the central axis **910**.

(61) As shown, an open channel may extend from the central passage **906** to the connection chamber **912**. The open channel may be open-ended on both ends. Alternatively, the connection chamber **912** may be closed-ended.

(62) As an example, the diameter of the central passage **906** may be 0.20". However, the diameter of the central passage **906** may be greater or less than 0.20".

(63) FIG. **14** illustrates a lateral view of a protuberance tether **1000** of a delivery catheter, according to an embodiment of the present disclosure. The protuberance tether **1000** may include a flexible tether **1002** having a primary feature, expanded end, or protuberance **1004** at a distal end **1006**. A proximal end **1007** of the tether **1002** may be operatively connected to a control device of an IMD delivery system, for example.

(64) The protuberance **1004** may be a sphere, block, pyramid, or various other such protuberances that provides an interfering feature within a central passage of an attachment member. The protuberance **1004** has a diameter **1008** that is less than the diameter of the central passage **906** (shown in FIGS. **12** and **13**). Accordingly, the protuberance **1004** may be sized and shaped to pass through the central passage **906**.

(65) As an example the diameter of the flexible tether **1002** may be 0.006", while the diameter of the protuberance may be 0.018". However, the diameter of the flexible tether **1002** may be greater or less than 0.006", while the diameter of the protuberance may be greater or less than 0.018".

(66) FIG. **15** illustrates a lateral view of a locking tether **1020** of a delivery catheter, according to an embodiment of the present disclosure. The locking tether **1020** may include a flexible tether **1022** similar to the tether **1002**. Notably, the locking tether **1020** does not include a protuberance at a distal end **1023**. The locking tether **1020** may be featureless, and may be devoid of a protuberance. For example, the locking tether **1020** may be featureless in that it may merely be an elongated strand of material, such as string, wire, or the like, having a uniform diameter throughout or a different diameters at particular areas. The distal end **1023** may provide or otherwise be part of an elongated interfering segment that is configured to provide a blocking or interfering barrier within the central passage **906** of the attachment member **900**. When disposed within the central passage **906**, the elongated interfering segment prevents the protuberance **1004** from passing into the central passage **906**. A proximal end **1024** of the tether **1022** may be operatively connected to a control device of an IMD delivery system, for example.

(67) The elongated interfering segment may be the same diameter as the rest of the tether **1022**, including the proximal end **1024**. Optionally, the elongated interfering segment may be outwardly flared, or tapered in relation to the remainder of the tether **1022**. Alternatively, a reduced-diameter extension may extend from a distal end of the elongated interfering segment. The reduced-diameter extension may have a diameter that is not great enough to block the protuberance **1008** from passing into and/or through the central passage **906**.

(68) As an example, the diameter of the flexible tether **1022** may be 0.006". However, the diameter of the flexible tether **1022** may be greater or less than 0.006".

(69) FIG. **16A** illustrates a cross-sectional view of the protuberance tether **1000** passing into the connection chamber **912** of the attachment member **900** of the IMD, according to an embodiment of the present disclosure. As shown, the protuberance **1008** fits within the central passage **906** and is able to pass therethrough. In order to securely tether the delivery catheter to the IMD, the locking tether **1020** is passed into the central passage **906**. The combined width of the locking tether **1020** and the protuberance **1008** within the central passage **906** and/or the connection chamber **912** is great enough to securely lodge the protuberance **1008** and the abutting portion of the locking tether

1020 within the central passage **906** and/or the connection chamber **912**, thereby securely tethering the IMD to the delivery catheter.

(70) FIG. **16B** illustrates an axial cross-sectional view of the protuberance **1008** within the central passage **906** of the attachment member **900**. It is to be understood that the view shown in FIG. **16B** is simplified and is not necessarily drawn to scale. While in the central passage **906**, a clearance area, gap, or the like **1009** is formed between an outer surface of the protuberance **1008** and an inner edge that defines the central passage **906**. The clearance area **1009** provides a diametric clearance between the protuberance **1008** and the central passage **906** that allows the protuberance **1008** to pass therethrough. The clearance area **1009** is generally less than a diameter of the elongated interfering segment of the locking tether **1020**. As such, when the elongated interfering segment of the locking tether **1020** is disposed within the central passage **906**, the protuberance **1008** is blocked from passing therein.

(71) FIG. **16C** illustrates an axial cross-sectional view of the protuberance **1008** within the central passage **906** of the attachment member **900**. It is to be understood that the view shown in FIG. **16C** is simplified and is not necessarily drawn to scale. As shown, the protuberance **1008** may abut one side of a wall or edge portion that defines the central passage **906**, but the total clearance area **1009** remains the same as shown in FIG. **16B**.

(72) FIG. **17** illustrates a cross-sectional view of the protuberance tether **1002** and the locking tether **1020** securely lodged within the connection chamber **912** of the attachment member **900** of an IMD, according to an embodiment of the present disclosure. As shown, the combined width of the protuberance **1008** and the abutting portion **1030** of the locking tether **1020** lodges the protuberance **1008** within the connection chamber **912**. As such, the tethers **1002** and **1020** are securely tethered to the attachment member **900**. In this manner, the IMD is securely tethered to the delivery catheter.

(73) As noted above, the locking tether **1020** may include a reduced-diameter extension, such as a string, that extends from a distal end of the elongated interfering segment **1023**. The extension may be sized and shaped so that the protuberance **1008** may dislodge out of the attachment member **900** even when the extension is within the central passage **906**.

(74) FIG. **18** illustrates a perspective bottom view of a protuberance **1008** of a protuberance tether **1002** and a locking tether **1020** securely lodged within the connection chamber **912** of the attachment member **900** of an IMD, according to an embodiment of the present disclosure. Because the locking tether **1020** is positioned within the central passage **906**, the protuberance **1008** is unable to pass through the central passage **906**, and is instead lodged within the connection chamber **912**. For example, because the combined diameter of the locking tether **1020** and the protuberance **1008** is greater than the diameter of the central passage **906**, the protuberance **1008** is unable to pass into the central passage **906** when the locking tether **1020** is positioned therein.

(75) Referring to FIGS. **17** and **18**, in order to release the IMD from the delivery catheter, the elongated interfering segment **1023** of the locking tether **1020** is removed from the central passage **906** in the direction of arrow **1040**. Once the locking tether **1020** is removed from the central passage **906**, the protuberance **1008** may fit through the central passage **906**. As such, the protuberance tether **1002** may also be removed from the central passage in the direction of arrow **1040**, thereby releasing the IMD from the delivery catheter.

(76) Continuing with the examples noted above, if the diameter of the central passage **906** is 0.020", and the diameter of the protuberance **1008** is 0.018", the protuberance **1008** is able to pass into the central passage **906**. However, when the interfering segment **1023** of the locking tether **1020** is positioned within the central passage **906**, the diameter of the interfering segment **1023** within the central passage **906** prevents the protuberance **1008** from passing into the central passage **906**. For example, the combined diameter of the protuberance **1008** and the interfering segment **1023** is 0.024", which is greater than the 0.020" diameter of the central passage **906**. Once the interfering segment **1023** is removed from the central passage **906**, such as by being retracted

into a delivery catheter, the protuberance **1008** may pop out of, or otherwise be removed from, the central passage **906**. Note, however, that a reduced diameter portion extending from the interfering segment **1023** may still be in the central passage **906** when the protuberance **1008** is removed from the central passage **906**. In this manner, the protuberance **1008** may be removed from the central passage **906** in response to the interfering segment **1023** of the locking tether **1020** being removed from the central passage **906**.

(77) FIG. **19** illustrates a flow chart of a method of implanting an IMD at an implant site, according to an embodiment of the present disclosure. The process begins at **1100**, in which a delivery catheter containing an IMD is guided to an implant site. Once at the implant site, the IMD is pushed out from the delivery catheter and fixed to the implant site at **1102**. Then, at **1104**, the delivery catheter is moved away from the implant site while remaining tethered to the IMD.

(78) When the IMD is fixed to the implant site and tethered to the delivery catheter, the IMD is then tested at **1106** to determine whether the IMD is properly physically and electrically connected to the implant site. At **1108**, it is determined whether the IMD is properly affixed to the implant site. If not, the method proceeds to **1110**, in which the IMD is removed from the implant site and retracted back into the delivery catheter and docked thereto. The process then returns to **1100**.

(79) If, however, the IMD is properly affixed to the implant site, the process continues to **1112**, in which a locking tether of a delivery catheter is removed from a central passage of an attachment member of the IMD. Then, at **1114**, a protuberance tether of the delivery catheter is removed from the central passage of the attachment member, thereby releasing the IMD from the delivery catheter.

(80) It has been found that the embodiments described with respect to FIGS. **12-19** are less susceptible to changes in length (compared to known systems) that would otherwise occur due to changes in temperature, deflection, torsion, and the like. Further, the embodiments described with respect to FIGS. **12-19** are not susceptible to inadvertent, spontaneous release, for example.

(81) FIG. **20** illustrates a simplified view of an IMD **1210** in electrical communication with at least three leads **1220**, **1224**, and **1230** implanted into a patient's heart **1212**, according to an embodiment. The IMD **1210** may be implanted into the heart **1212** and released from a delivery catheter, such as described above with respect FIGS. **1-19**.

(82) To sense atrial cardiac signals and to provide right atrial chamber stimulation therapy, the IMD **1210** may be coupled to an implantable right atrial lead **1220** including at least one atrial tip electrode **1222** that typically is implanted in the patient's right atrial appendage. The right atrial lead **1220** may also include an atrial ring electrode **1223** to allow bipolar stimulation or sensing in combination with the atrial tip electrode **1222**.

(83) To sense the left atrial and left ventricular cardiac signals and to provide left-chamber stimulation therapy, the IMD **1210** may be coupled to a lead **1224** designed for placement in the "coronary sinus region" via the coronary sinus ostium in order to place a distal electrode adjacent to the left ventricle and additional electrode(s) adjacent to the left atrium. As used herein, the phrase "coronary sinus region" refers to the venous vasculature of the left ventricle, including any portion of the coronary sinus, great cardiac vein, left marginal vein, left posterior ventricular vein, middle cardiac vein, and/or small cardiac vein or any other cardiac vein accessible by the coronary sinus.

(84) Accordingly, the lead **1224** is designed to: receive atrial and/or ventricular cardiac signals; deliver left ventricular pacing therapy using at least one left ventricular tip electrode **1226** for unipolar configurations or in combination with left ventricular ring electrode **1225** for bipolar configurations; deliver left atrial pacing therapy using at least one left atrial ring electrode **1227** as well as shocking therapy using at least one left atrial coil electrode **1228**.

(85) The IMD **1210** is also shown in electrical communication with the patient's heart **1212** by way of an implantable right ventricular lead **1230** including, in the embodiment, a right ventricular (RV) tip electrode **1232**, a right ventricular ring electrode **1234**, a right ventricular coil electrode **1236**, a

superior vena cava (SVC) coil electrode **1238**, and so on. Typically, the right ventricular lead **1230** is inserted transvenously into the heart **1212** so as to place the right ventricular tip electrode **1232** in the right ventricular apex such that the RV coil electrode **1236** is positioned in the right ventricle and the SVC coil electrode **1238** will be positioned in the right atrium and/or superior vena cava. Accordingly, the right ventricular lead **1230** is capable of receiving cardiac signals, and delivering stimulation in the form of pacing and shock therapy to the right ventricle.

(86) The IMD **1210** may be one of various types of implantable devices, such as, for example, an implantable pacemaker, implantable cardioverter-defibrillator (“ICD”), defibrillator, cardiac rhythm management (“CRM”) device, neurostimulator, electrophysiology (“EP”) mapping and radio frequency (“RF”) ablation system, or the like.

(87) While various spatial and directional terms, such as top, bottom, lower, mid, lateral, horizontal, vertical, front and the like may be used to describe embodiments of the present disclosure, it is understood that such terms are merely used with respect to the orientations shown in the drawings. The orientations may be inverted, rotated, or otherwise changed, such that an upper portion is a lower portion, and vice versa, horizontal becomes vertical, and the like.

(88) It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the disclosure without departing from its scope. While the dimensions, types of materials and coatings described herein are intended to define the parameters of the disclosure, they are by no means limiting and are exemplary embodiments. Many other embodiments will be apparent to those of skill in the art upon reviewing the above description. The scope of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means—plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

Claims

1. A delivery system, comprising: a control device; and a plurality of tethers operatively coupled to the control device, wherein the plurality of tethers are configured to fit through a central passage of an implantable medical device (IMD) in a first mode, wherein the plurality of tethers are configured to not fit through the central passage in a second mode, and wherein the delivery system is configured to test whether the IMD is connected to an implant site in the second mode.
2. The delivery system of claim 1, wherein the plurality of tethers include a locking tether having a protuberance at a distal end of a tether.
3. The delivery system of claim 2, wherein the protuberance has a larger diameter than the tether.
4. The delivery system of claim 1, wherein the plurality of tethers include a locking tether and a flexible tether, and wherein the locking tether is locked in a connection chamber of the IMD in the second mode.
5. The delivery system of claim 4, wherein a first distal end of the flexible tether is proximal to a second distal end of the locking tether in the first mode.
6. The delivery system of claim 5, wherein the first distal end is distal to the second distal end in the second mode.
7. An implantable medical device (IMD) system, comprising: an IMD including an attachment

member having a central passage; and a delivery system including a plurality of tethers operatively coupled to a control device, wherein the plurality of tethers are configured to fit through the central passage in a first mode, wherein the plurality of tethers are configured to not fit through the central passage in a second mode, and wherein the delivery system is configured to test whether the IMD is connected to an implant site in the second mode.

8. The IMD system of claim 7, wherein the plurality of tethers include a locking tether having a protuberance at a distal end of a tether.

9. The IMD system of claim 8, wherein the protuberance has a larger diameter than the tether.

10. The IMD system of claim 7, wherein the plurality of tethers include a locking tether and a flexible tether, and wherein the locking tether is locked in a connection chamber of the IMD in the second mode.

11. The IMD system of claim 10, wherein a first distal end of the flexible tether is proximal to a second distal end of the locking tether in the first mode.

12. The IMD system of claim 11, wherein the first distal end is distal to the second distal end in the second mode.

13. A method, comprising: advancing an implantable medical device (IMD) system to an implant site, wherein the IMD system includes a delivery system tethered to an IMD; affix the IMD to the implant site; remove the delivery system from the implant site, wherein the delivery system remains tethered to the IMD in a mode after removal; and test whether the IMD is connected to the implant site in the mode.

14. The method of claim 13, wherein testing whether the IMD is connected to the implant site includes testing whether the IMD is physically connected to the implant site.

15. The method of claim 13, wherein testing whether the IMD is connected to the implant site includes testing whether the IMD is electrically connected to the implant site.

16. The method of claim 13 further comprising removing, in response to the IMD being connected to the implant site, a plurality of tethers from a central passage of the IMD.

17. The method of claim 13 further comprising removing, in response to the IMD not being connected to the implant site, the IMD from the implant site.

18. The method of claim 13, wherein the delivery system is tethered to the IMD by a plurality of tethers including a locking tether having a protuberance at a distal end of a tether, and wherein the protuberance has a larger diameter than the tether.

19. The method of claim 13, wherein the delivery system is tethered to the IMD by a plurality of tethers including a locking tether and a flexible tether, and wherein the locking tether is locked in a connection chamber of the IMD in the second mode.

20. The method of claim 19, wherein the first distal end is distal to the second distal end in the second mode.
